

Online Product Reviews Sentimental Analysis



- ❖ **Model Selection**
- ❖ **Methodology**
- ❖ **What we did**
- ❖ **Timeline**

CONTENTS

Model selection :

In this study, an attempt has been made to classify sentiment analysis for product reviews using machine learning techniques. Two algorithms namely support vector machine and Random forest gave best results

By considering three ML Evaluation metrics used here are accuracy, precision and recall.

Using random forest,

Accuracy of prediction is 95.0%.
Precision of prediction is 0.95.
Recall of prediction is 0.94.

Using Support vector machine,

Accuracy of prediction is 92.166%.
Precision of prediction is 0.92.
Recall of prediction is 0.91.

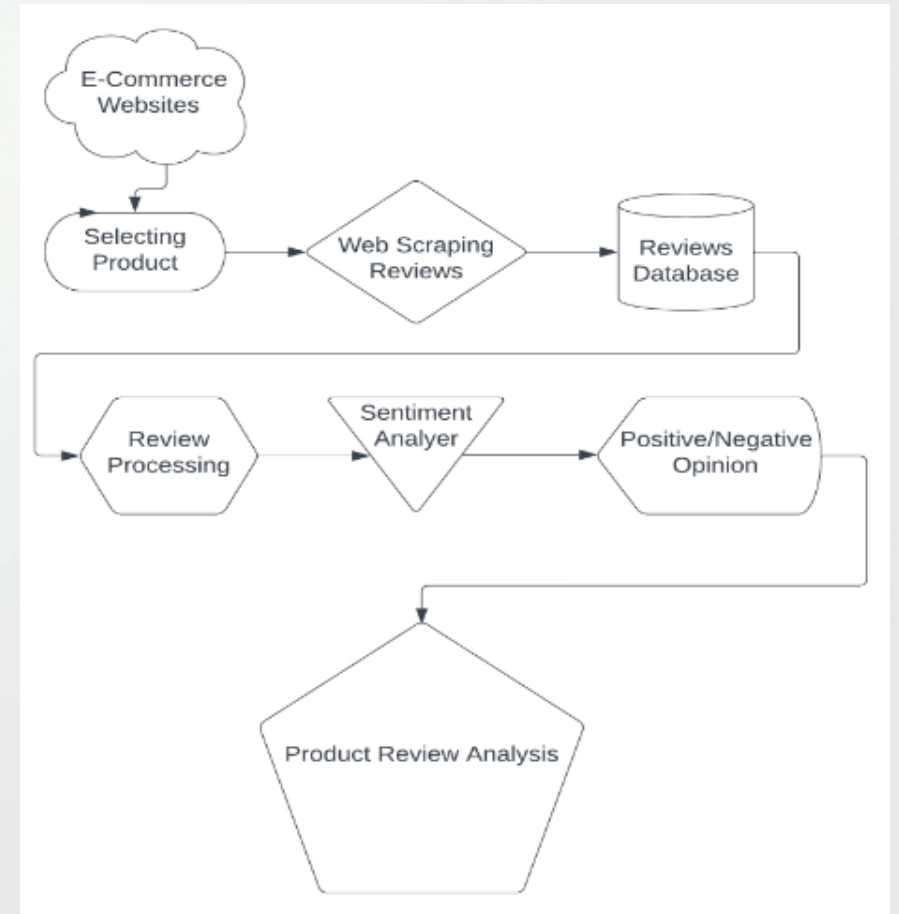
Using Logistic regression,

Accuracy of prediction is 91.2%.
Precision of prediction is 0.91.
Recall of prediction is 0.90.

From the above results, support vector machine is slightly better method compared to Logistic regression and By using Random forest, we obtained 95.0% accuracy. Hence Random forest was used to built our model which means the model built for the prediction of sentiment of the product review gives about 95% right prediction.

Methodology

- BeautifulSoup is used for webscraping
- Libraries used- numpPy , pandas, matplotlib, sklearn
- Data cleaning is done by Lowercasing , Removing Puncuatations, Removing Numbers, Removing extra space, Replacing the repetitions of punctations, Removing Emojis.
- All NLTK classifiers work with feature structures, which can be simple dictionaries mapping a feature name to a feature value.
- We have made a comparative study of the performance of 4 Machine Learning classifier models
- We have used flask for front-end of our project
- Flask framework is used to develop web applications using python for built-in development server.



What we did ?

Web Scrapping

- ❑ First we have performed web scrapping in online shopping websites like Amazon and flipkart.

Data Cleaning

- ❑ We examined all the columns and removed unnecessary data and duplicates in order to get unbiased results for the analysis of the data.

Text Preprocessing

- ❑ Stemming, stop-word removal, removing the html tags, punctuations and Lemmatization.

Model Selection

- ❑ We did preprocessing to remove insignificant columns. Then We have performed model selection.

Model Training

- ❑ we divided data into testing(20%) and training(80%) data and then built the model using Random forest

Tested Model

- ❑ Finally we tested the model we built on the testing data.

What we did ?

In [24]:

```
product_names
'Samsung Galaxy S20 FE 5G (Cloud Navy, 8GB RAM, 128GB Storage) with No Cost EMI & Additional Exchange Offers',
'Samsung Galaxy M13 5G (Midnight Blue, 4GB, 64GB Storage) | 5000mAh Battery | Upto 8GB RAM with RAM Plus',
'Samsung Galaxy M13 (Midnight Blue, 6GB, 128GB Storage) | 6000mAh Battery | Upto 12GB RAM with RAM Plus',
'Samsung Galaxy A13 Peach, 6GB RAM, 128GB Storage with No Cost EMI/Additional Exchange Offers (SM-A135FZQJINS)',
'Samsung Galaxy M32 5G (Slate Black, 8GB RAM, 128GB Storage) | Dimensity 720 Processor | 5000mAh Battery| Knox Security',
'Samsung Galaxy A13 Black, 4GB RAM, 64GB Storage with No Cost EMI/Additional Exchange Offers, (SM-A135FZKJINS)',
'Samsung Galaxy A23 Peach, 6GB RAM, 128GB Storage with No Cost EMI/Additional Exchange Offers',
'Samsung Galaxy M04 Dark Blue, 4GB RAM, 128GB Storage | Upto 8GB RAM with RAM Plus | MediaTek Helio P35 | 5000 mAh Battery',
'iQOO 9 Pro 5G (Legend, 8GB RAM, 256GB Storage) | Snapdragon 8 Gen 1 Mobile Processor | 120W FlashCharge',
'Samsung Galaxy A23 Peach, 6GB RAM, 128GB Storage with No Cost EMI/Additional Exchange Offers',
'Samsung Galaxy M52 5G (ICY Blue, 8GB RAM, 128GB Storage) Latest Snapdragon 778G 5G | sAMOLED 120Hz Display',
'Samsung Galaxy A13 Blue, 6GB RAM, 128GB Storage with No Cost EMI/Additional Exchange Offers, (SM-A135FLBJINS)',
'Samsung Galaxy M13 (Stardust Brown, 4GB, 64GB Storage) | 6000mAh Battery | Upto 8GB RAM with RAM Plus',
'Samsung Galaxy A33 5G Orange, 6GB RAM, 128GB Storage',
'Samsung Galaxy M53 5G (Deep Ocean Blue, 6GB, 128GB Storage) | 108MP | sAmoled+ 120Hz | 12GB RAM with RAM Plus | Travel Adapter to be Purchased Separately',
'Samsung Galaxy A33 5G Black, 8GB RAM, 128GB Storage',
'Samsung Galaxy A13 Blue, 4GB RAM, 64GB Storage with No Cost EMI/Additional Exchange Offers, (SM-A135FLBJINS)',
'Samsung Galaxy A33 5G Black, 6GB RAM, 128GB Storage',
'Samsung Galaxy A33 5G Black, 8GB RAM, 128GB Storage'
```

In [25]:

```
len(product_names)
```

Out[25]: 324

In [67]:

```
reviews=[]
ratings=[]
title=[]
starNum = ['one_star','two_star','three_star','four_star','five_star']
# 50 X 10 = 500 reviews => 5X10pages(each 10reviews) => 10,000
for j in range(len(link)):
    for k in range(len(starNum)): #<= give no. of pages #(10reviews per page) ### 5(star ratings) X no. of pages X 10(reviews/page)
        ## 1(no. of pages) X 10(reviews per page) X 5(no. of stars) X 100(no. of products) =
        response=Searchreviews(link[j]+'&pageNumber='+str(k)+'&filterByStars='+starNum[k])
        soup=BeautifulSoup(response.content)
        for i in soup.findAll("span",{'data-hook':"review-body"}):
            reviews.append(i.text)
        for p in soup.findAll("i",{'data-hook':"review-star-rating"}):
            ratings.append(p.findAll("span")[0].contents[0])
        for t in soup.findAll("a",{'data-hook':"review-title"}):
            title.append(t.text)
```

```
https://www.amazon.in/Samsung-Emerald-Storage-Purchased-Separately/product-reviews/B0814WR9L1/ref=cm_cr_dp_d_show_all_btm?ie=UTF8&reviewerType=all_reviews&pageNumber=28&filterByStar=three_star
https://www.amazon.in/Samsung-Emerald-Storage-Purchased-Separately/product-reviews/B0814WR9L1/ref=cm_cr_dp_d_show_all_btm?ie=UTF8&reviewerType=all_reviews&pageNumber=38&filterByStar=four_star
https://www.amazon.in/Samsung-Emerald-Storage-Purchased-Separately/product-reviews/B0814WR9L1/ref=cm_cr_dp_d_show_all_btm?ie=UTF8&reviewerType=all_reviews&pageNumber=48&filterByStar=five_star
https://www.amazon.in/Samsung-Galaxy-Prime-Light-Blue/product-reviews/B08D3T6Z1D/ref=cm_cr_dp_d_show_all_btm?ie=UTF8&reviewerType=all_reviews&pageNumber=08&filterByStar=one_star
https://www.amazon.in/Samsung-Galaxy-Prime-Light-Blue/product-reviews/B08D3T6Z1D/ref=cm_cr_dp_d_show_all_btm?ie=UTF8&reviewerType=all_reviews&pageNumber=18&filterByStar=two_star
https://www.amazon.in/Samsung-Galaxy-Prime-Light-Blue/product-reviews/B08D3T6Z1D/ref=cm_cr_dp_d_show_all_btm?ie=UTF8&reviewerType=all_reviews&pageNumber=28&filterByStar=three_star
https://www.amazon.in/Samsung-Galaxy-Prime-Light-Blue/product-reviews/B08D3T6Z1D/ref=cm_cr_dp_d_show_all_btm?ie=UTF8&reviewerType=all_reviews&pageNumber=38&filterByStar=four_star
https://www.amazon.in/Samsung-Galaxy-Prime-Light-Blue/product-reviews/B08D3T6Z1D/ref=cm_cr_dp_d_show_all_btm?ie=UTF8&reviewerType=all_reviews&pageNumber=48&filterByStar=five_star
https://www.amazon.in/Samsung-Galaxy-Storage-Additional-Exchange/product-reviews/B09VK508XK/ref=cm_cr_dp_d_show_all_btm?ie=UTF8&reviewerType=all_reviews&pageNumber=08&filterByStar=one_star
https://www.amazon.in/Samsung-Galaxy-Storage-Additional-Exchange/product-reviews/B09VK508XK/ref=cm_cr_dp_d_show_all_btm?ie=UTF8&reviewerType=all_reviews&pageNumber=18&filterByStar=two_star
```

In [73]:

```
rev={'titles':title,'reviews':reviews,'ratings': ratings}
rev
```

```
"\nDon't waste your money in this phone.\n",
'\nWorst mobile\n',
'\nSamsung M13 5G Phone is not Good. Every thing is slow .\n',
'\nmediocre laggy device\n',
'\nGood Buget Smart Phone But *Very Slow*\n',
'\nPhoto hanging all time\n',
'\nFeatures are not as expected\n',
'\nPerformance\n',
'\nInternet and system\n',
'\nVery slow and oversized phone.\n',
'\nNo sensation in vicinity, no gesture emergency light on like Motorola, rather slow ...'\n',
'\nCamera is very poor\n',
'\nGood\n',
'\nBest for battery\n',
'\nVery good mobile\n',
'\nNot some basic features\n',
'\nAverage quality\n',
'\nOkey according to price\n',
'\nCAUTION:PLS READ ND THEN THINK TO BUY\n',
'\nNot sure why phone is not getting charged.\n'
```

In [86]:

```
search_name
```

Out[86]: 'latest_samsung_mobiles'

In [87]:

```
review_data.to_csv(search_name+'_reviews.csv',index=False)
```

In [88]:

```
review=pd.DataFrame(review_data.groupby('ratings').size().sort_values(ascending=False).rename('No of Users').reset_index())
review.head()
```

Out[88]:

	ratings	No of Users
0	1.0 out of 5 stars	793
1	2.0 out of 5 stars	633
2	5.0 out of 5 stars	593
3	3.0 out of 5 stars	491
4	4.0 out of 5 stars	490

Results :

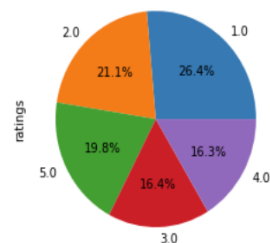
```
dataset['ratings'].value_counts()
```

1.0	793
2.0	633
5.0	593
3.0	491
4.0	490

```
Name: ratings, dtype: int64
```

```
dataset['ratings'].value_counts().plot(kind='pie', autopct='%1.1f%%')
```

```
<AxesSubplot:ylabel='ratings'>
```



```
In [18]: data1=pd.concat([dataset,dt],axis=1)
          data1.head()
```

Out[18]:

	titles	reviews	ratings	emotion
0	'nWorst service provided by Amazon Don't buy t... 'nPlease , Don't buy this product "samsung gal...		1.0	-1
1	'nNot up to the standard'in 'nordered on 12/10/2022 compared to old Samsun...		1.0	-1
2	'nWe are ditched'in 'nVery poor Quality phone has been sent to us ...		1.0	-1
3	'nDON'T BUY THIS PHONE'in 'nBought it after seeing 4 star ratings and go...		1.0	-1
4	'nToo slow processor'in 'nI want this product for normal use...But t...		1.0	-1

```
In [76]: review_data.replace('title', np.nan, inplace=True)
review_data.replace('reviews', np.nan, inplace=True)
review_data.replace('ratings', np.nan, inplace=True)
```

```
In [77]: review_data.sample(5, replace=True)
```

Out[77]:

	titles	reviews	ratings
2127	iPhone look like new one in	InFirst of all, the phone's screen is NOT with...	4.0 out of 5 stars
1875	InThis product is very bed west of money/in	InMe is product to use kar raha hu na isme cha...	1.0 out of 5 stars
1758	InGood in	InGood product in	5.0 out of 5 stars
2552	नयाबरो बेकार इसकी खानि वारंटी में होते हुये	InMy phone is getting switched off while foldi...	5.0 out of 5 stars
1358	InPoor battery backup in	InThe phone is okay, but for a phone of 6.6 in...	2.0 out of 5 stars

```
In [55]: review_data.dropna(subset=['titles'], inplace=True)
review_data.dropna(subset=['reviews'], inplace=True)
review_data.dropna(subset=['ratings'], inplace=True)
```

```
In [27]: import string
def text_clean(Review):
    Review=Review.lower()
    Review=re.sub('[\.\?@]', '', Review)
    Review=re.sub('[%s]' % re.escape(string.punctuation), '', Review)
    Review=re.sub('n\+', '', Review)
    Review=re.sub('\n*\d*\n*', '', Review)
    Review=re.sub('[\'\"\"...]', '', Review)
    Review=Review.strip()
    Review=re.sub('\n', '', Review)
    return Review
cleaned=lambda x:text_clean(x)
```

```
In [28]: data2['cleaned_description']=pd.DataFrame(data2.Review.apply(cleaned))
data2.head(10)
```

Out[28]:

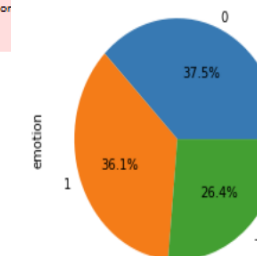
emotion	Review	cleaned_description
0	-1 \nWorst service provided by Amazon Don't buy t...	worst service provided by amazon dont buy this...
1	-1 \nNot up to the standard\nInordered on 12/10/...	not up to the standard ordered on compared to...
2	-1 \nWe are ditched\nInVery poor Quality phone h...	we are ditched very poor quality phone has bee...
3	-1 \nDON'T BUY THIS PHONE\nInbought it after see...	dont buy this phone bought it after seeing st...
4	-1 \nToo slow processor\nInI want this product f...	too slow processor I want this product for nor...
5	-1 \nNot worth the hype and money - Samsung shoul...	not worth the hype and money samsung should l...
6	-1 \nThe M series phone have software. \nInThe M...	the m series phone have software the m aeries...
7	-1 \nBad display Contacts app is very pathetic\nIn...	bad display contacts app is very pathetic disp...
8	-1 \nDon't waste your money in this phone. \nInDo...	dont waste your money in this phone dont waste...
9	-1 \nWorst mobile\nInThe samsung and Amazon both...	worst mobile the samsung and amazon both fool...

```
data2['emotion'].value counts()
```

```
0    1124
1    1083
-1     793
Name: emotion, dtype: int64
```

```
data2['emotion'].value_counts().plot(kind='pie', autopct='%1.1f%%')
```

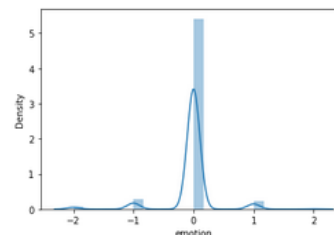
```
<AxesSubplot:ylabel='emotion'>
```



```
In [44]: import matplotlib.pyplot as plt
sns.distplot(y_test_pred)
plt.show()
```

C:\Users\DELL\anaconda3\lib\site-packages\seaborn\distributions.py:2619: FutureWarning: 'distplot' is a deprecated alias for the 'displot' function. Please adapt your code to use either 'displot' (a figure-level function) or 'histplot' (an axes-level function for histograms).

warnings.warn(msg, FutureWarning)



Results :

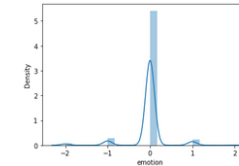
- By successfully implementing on our data we got an accuracy of about 95.0%

```
In [38]: from sklearn.pipeline import Pipeline
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
classifier=MultinomialNB()
tvec=TfidfVectorizer()
clf = MultinomialNB()
classifier=Pipeline([('vectorizer',tvec),('classifier',clf)])
classifier.fit(X_train, y_train)
```

```
In [39]: from sklearn.pipeline import Pipeline
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.ensemble import RandomForestClassifier
clf= RandomForestClassifier(n_estimators=10, criterion="entropy")
tvec=TfidfVectorizer()
classifier=Pipeline([('vectorizer',tvec),('classifier',clf)])
classifier.fit(X_train, y_train)
```

```
In [44]: import matplotlib.pyplot as plt
sns.distplot(y_test_pred)
plt.show()
```

C:\Users\OELL\anaconda3\lib\site-packages\seaborn\distributions.py:2619: FutureWarning: 'distplot' is a deprecated function and will be removed in a future version. Please adapt your code to use either 'displot' (a figure-level function with similar flexibility) or 'histplot' (an axes-level function for histograms).
warnings.warn(msg, FutureWarning)



- Then we created confusion matrix to check with the counts.

	precision	recall	f1-score	support
-1	0.97	0.89	0.93	157
0	0.94	0.96	0.95	223
1	0.95	0.98	0.96	220
accuracy			0.95	600
macro avg	0.95	0.94	0.95	600
weighted avg	0.95	0.95	0.95	600

Confusion Matrix :

```
[[140  11   6]
 [  3 214   6]
 [  1   3 216]]
```

Accuracy: 95.0

What we did ?

```
**Testing the model

In [45]: text = ["waste product"]
yp = classifier.predict(text)
print(yp)

[-1]

In [46]: text = ["I Love this Product, quality is good"]
yp = classifier.predict(text)
print(yp)

[1]

In [47]: text = ["It seems 1 size small ... please order 1 size bigger.....so if you are 8 let say order 9.... because this shoes is very
yp = classifier.predict(text)
print(yp)

[-1]

In [48]: < ok not greaCamera quality is not good selfie camera is worst if you are buying this phone for camera you will be disappointed"]

[0]

In [49]: text = ['The camera is like the worst ever any phone has released for 20k price range. Even though they say it has 108mp cam the p
yp = classifier.predict(text)
print(yp)

[-1]
```

Activate
Go to Settings

What we did ?

```
File Edit Selection View Go Run Terminal Help
new_mod.html - cp - Visual Studio Code

new_mod.html x flask_app.py
templates > new_mod.html > html > style > left
1
2 <!(DOCTYPE html)
3 <html lang="en">
4 <script language="javascript" type="text/javascript">
5
6 function onlyAlphabets(e, t) {
7   try {
8     if (window.event) {
9       var charCode = window.event.keyCode;
10    }
11    else if (e) {
12      var charCode = e.which;
13    }
14    else { return true; }
15    if ((charCode > 64 && charCode < 91) || (charCode > 96 && charCode < 123))
16      return true;
17    else
18      return false;
19  }
20  catch (err) {
21    alert(err.Description);
22  }
23 }
24
25 </script>
26
27 * { box-sizing: border-box; }
28 @import url('https://fonts.googleapis.com/css?family=Rubik:400,500&display=swap');
29
30 body {
31   font-family: 'Rubik', sans-serif;
32   /* background-color: #eee; */
33 }
34
35
36 .container {
37   display: flex;
38   height: 100vh;
39 }
40
41 .left {
42   overflow: hidden;
43   display: flex;
44   flex-wrap: wrap;
```

```
File Edit Selection View Go Run Terminal Help
flask_app.py - cp - Visual Studio Code

new_mod.html x flask_app.py x
flask_app.py > login
1 import numpy as np
2 import re
3 import nltk
4 from flask import Flask, render_template, request
5 import pickle
6
7 # import pickle
8 import os
9
10 # import TextBlob
11 from textblob import TextBlob
12
13 from nltk.stem.porter import PorterStemmer
14 ps = PorterStemmer()
15
16 PHOTOS_FOLDER = os.path.join('/', 'static', 'photos')
17
18 # /static/photos/Positive.gif
19 app = Flask(__name__)
20
21 app.config['UPLOAD_FOLDER'] = PHOTOS_FOLDER
22 model = pickle.load(open('119.pkl', 'rb'))
23
24 @app.route('/')
25 def home():
26     return render_template("new_mod.html")
27
28 @app.route('/login', methods = ['POST'])
29 def login():
30     text = request.form["review"]
31     tempy = TextBlob(text)
32     sample = tempy.correct()
33
34     total = [str(sample)]
35     y_pred = model.predict(total)
36
37     temp = "Given Text : "
38
39     if(y_pred == 1):
40         text3 = os.path.join(app.config['UPLOAD_FOLDER'], 'PosMochi.gif')
41         text2 = "POSITIVE REVIEW"
42         colr = "colgren"
43     elif(y_pred == 0):
44         text3 = os.path.join(app.config['UPLOAD_FOLDER'], 'neutral.gif')
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE

```
PS D:\capstone_project\cp> & C:/Users/Lenovo/AppData/Local/Programs/Python/Python311/python.exe d:/capstone_project/cp/flask_app.py
d:\capstone_project\cp\flask_app.py:22: DeprecationWarning: Please use `csr_matrix` from the `scipy.sparse` namespace, the `scipy.sparse.csr` namespace is deprecated.
  model = pickle.load(open('119.pkl', 'rb'))
* Serving Flask app 'flask_app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
d:\capstone_project\cp\flask_app.py:22: DeprecationWarning: Please use `csr_matrix` from the `scipy.sparse` namespace, the `scipy.sparse.csr` namespace is deprecated.
  model = pickle.load(open('119.pkl', 'rb'))
* Debugger is active!
* Debug PIN: 405-601-399
```

Review Analysis

Enter your Review

Predict



Given Text :

great deal

POSITIVE REVIEW

Activate Windows
Go to Settings to activate Windows.

Review Analysis

Enter your Review

Predict



Given Text :

bad product received

NEGATIVE REVIEW

Activate Windows
Go to Settings to activate Windows.

Review Analysis

Update -
Upgrading from 1 star to 3 star after
issues is rectified.
The speaker issue was rectified after

Predict



Given Text :

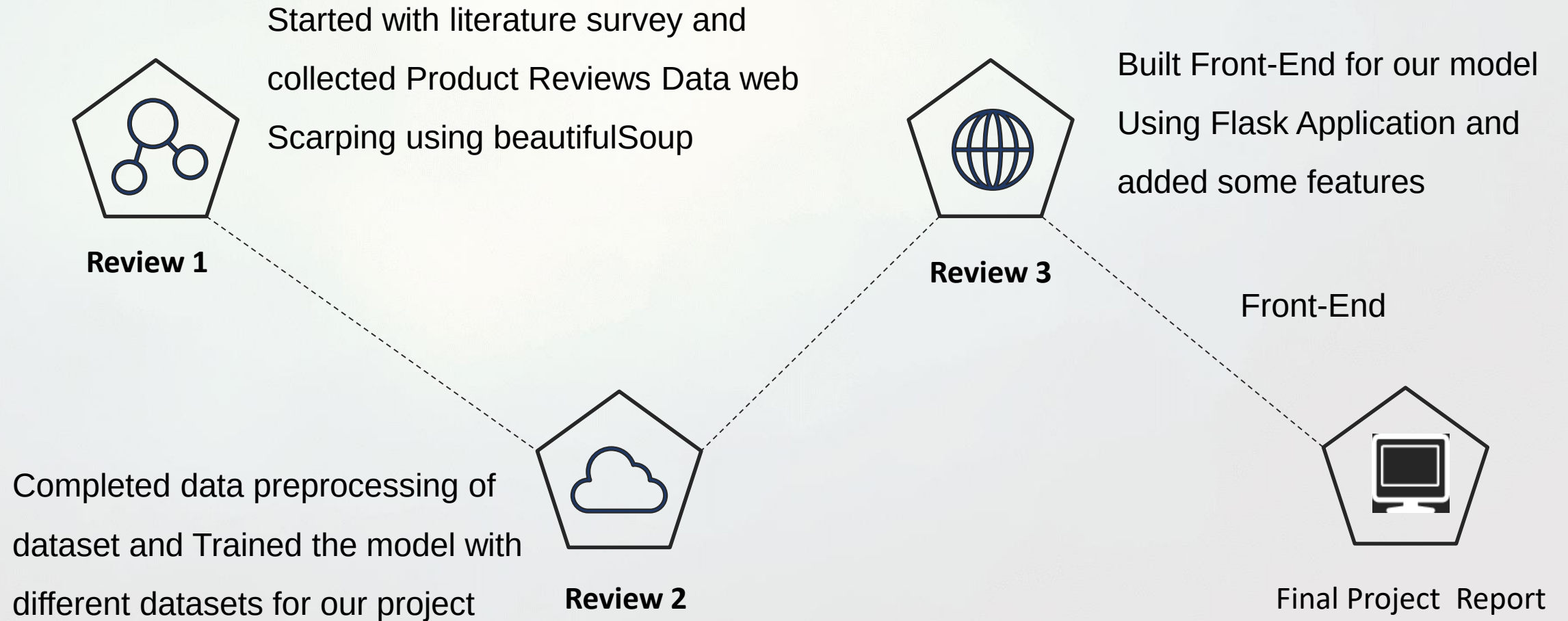
I bought Samsung S20 mobile phone on end of the July'22. after 4th month using this phone on 1st week of Dec'22 suddenly phone was stoped working. I am trying to reach Amazon customer care for returning the mobile but replacement is happen only within 14 days of purchase. Same day i went Samsung customer care center @ Iyandagar Bangalore for checking what's wrong with the phone. they

NEUTRAL REVIEW

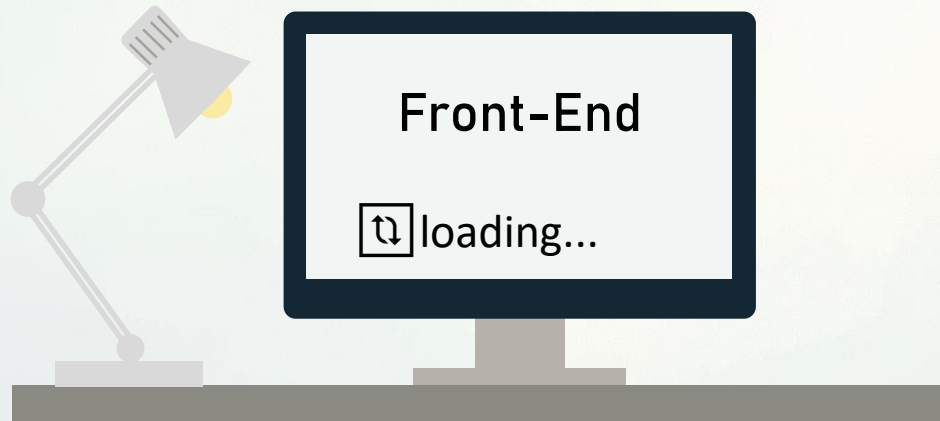
Activate Windows

Go to Settings to activate Windows.

Time Plan :

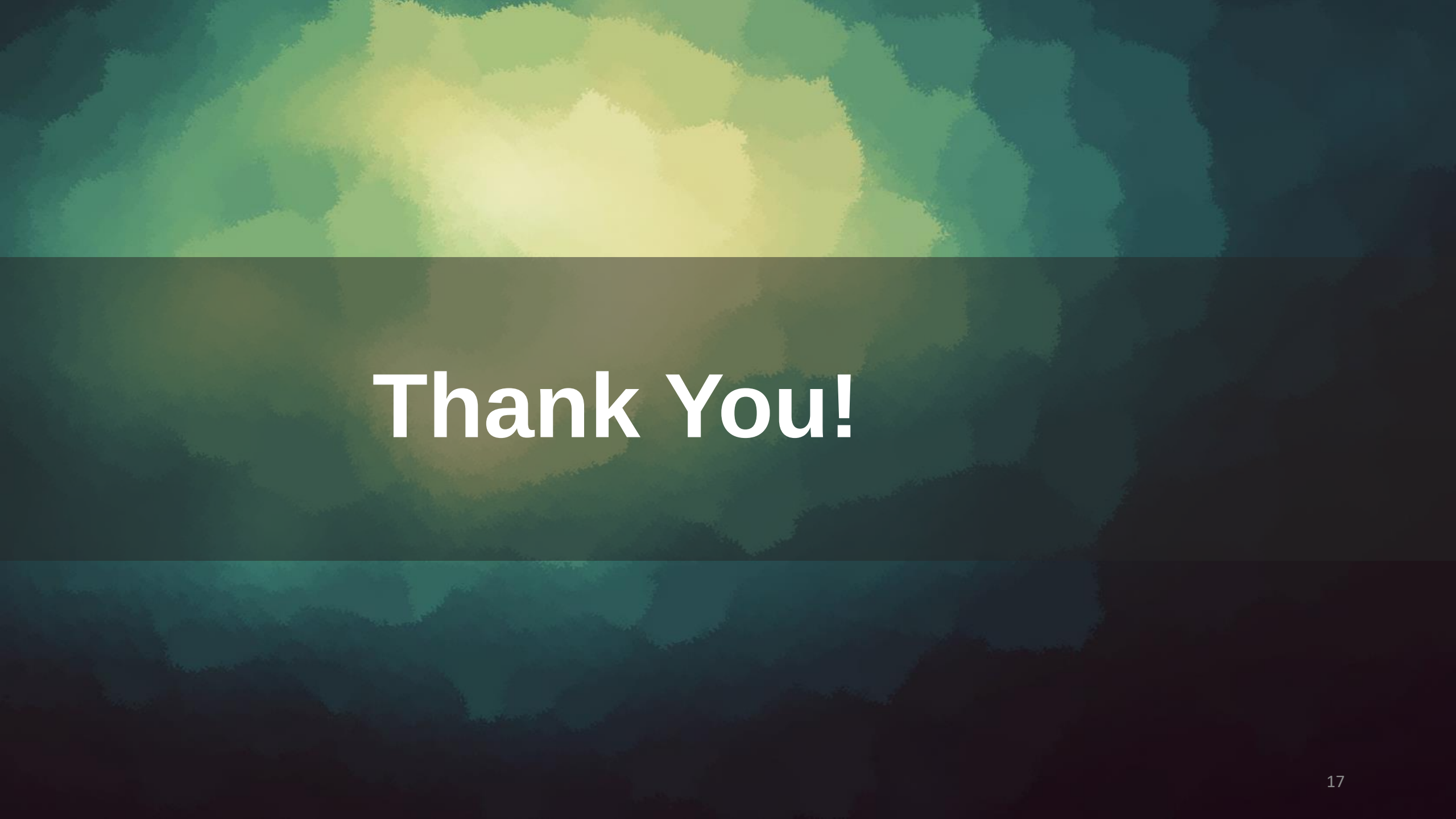


What we do :



Features for Review 3

- ▶ For review-2 We have trained the model using best fit model and created a home page for our project in front end.
- ▶ Still need to do the changes in app.py(Flask Application file) and add some features by our next Review-3.

The background features a vibrant, abstract design. The top half is filled with soft, painterly clouds in shades of lime green and pale yellow. The bottom half is a dark teal gradient. A horizontal band of dark teal color spans the width of the slide, serving as a backdrop for the text.

Thank You!